CarDiff

Anatomically-Aware Dental Caries Synthesis using Diffusion

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Scan: paper & code

1 Background & Motivation

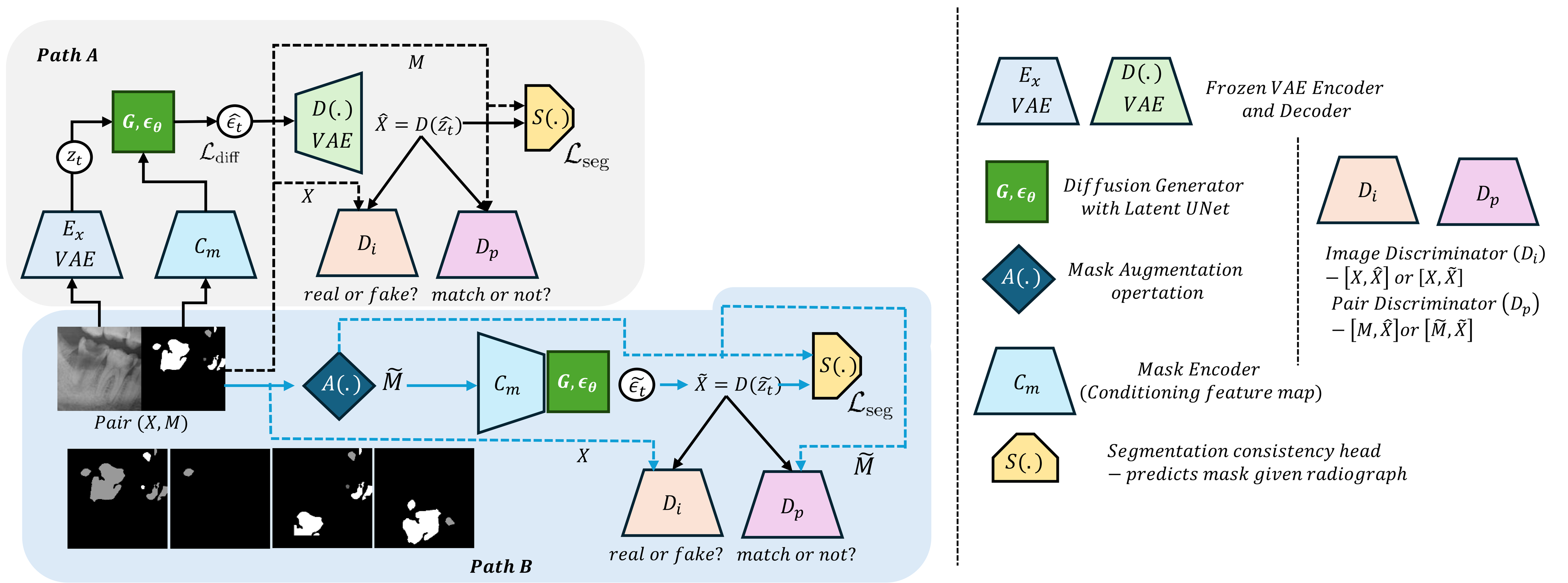
- Caries analysis is limited by **scarce annotations** and **class imbalance**.
- **Deep** caries dominate; **superficial** & **medium** caries are rare.
- Pixel-space generators ignore **anatomy** & **pathology**, giving implausible radiographs.

In MLUA-DC1000, deep caries make up **67%** of lesions versus only **21%** medium and **12%** superficial — so minority classes are segmented poorly. **Goal:** anatomy-aware synthesis to *augment* rare caries classes.

2 Contributions

- **CarDiff** — a semi-supervised, segmentation-guided latent diffusion framework for dental caries synthesis.
- **Dual-path training**: fully paired mask–image learning *(A)* plus self-supervised generation from augmented masks *(B)*.
- **Causal conditioning**: patch-GNN + global attention, injected via **FiLM**.
- Best FID & LPIPS vs. 4 baselines; downstream Dice gains across **7** segmentation nets.

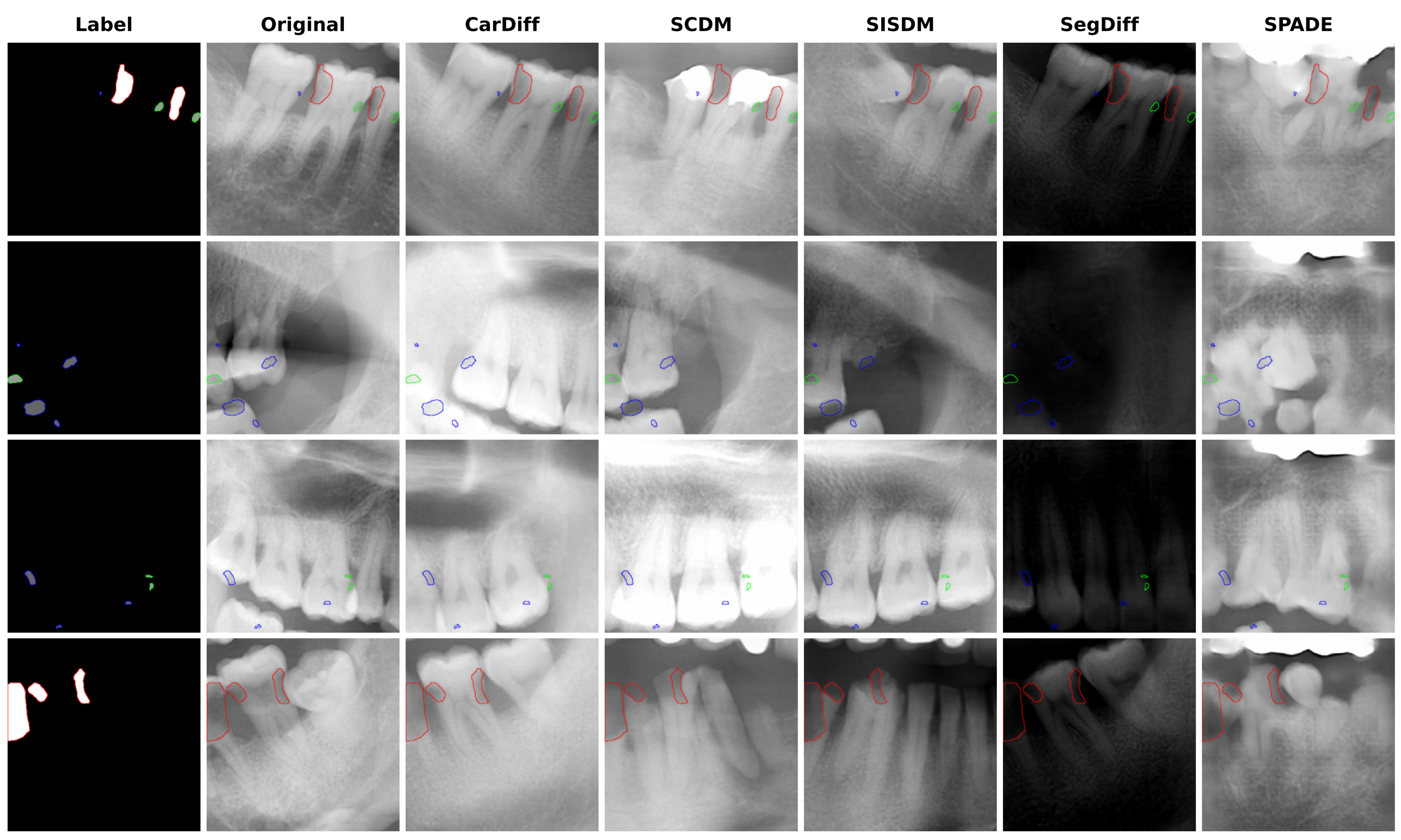
3 The CarDiff Framework



- **Path A** (paired): frozen-VAE latent; generator denoises given mask $C_m(M)$.
- **Path B** (self-sup.): synthesize from augmented masks $\tilde{M}=A(M)$ — no paired image.
- **Consistency** head $S(\cdot)$ + discriminators D_i, D_p enforce realism & mask–image match.

Losses: diffusion + Dice seg-consistency + adversarial.

4 Qualitative Comparison



CarDiff → **realistic attenuation** & **sharper boundaries**.

Boundaries: **blue** = superficial, **green** = medium, **red** = deep.

5 Dataset & Experimental Setup

- **Data:** MLUA-DC1000 — **2615** RoIs (384^2), split **1833** / **521** / **261**.
- **Setup:** latent diffusion (1000 steps); 7 nets, ResNet-34, Dice, 50 epochs.

6 Results: Synthesis & Augmentation

Synthesis quality

Model	FID↓	LPIPS↓
CarDiff	**90.8**	**0.48**
SCDM	95.6	0.51
SISDM	100.0	0.51
SegDiff	126.7	0.64
SPADE	164.1	0.52

Best FID & LPIPS — 5% / 6.6% over next best.

Dice: Base vs. +Aug (5×)

Arch.	Base	+Aug
U-Net	0.40	**0.41**
U-Net++	0.37	**0.41**
FPN	0.34	**0.40**
LinkNet	0.37	**0.42**
MA-Net	0.38	0.38
PAN	**0.40**	0.38
PSPNet	0.35	**0.39**

Up to **+6.5%** Dice; **MC +25.7%**.

7 Conclusions

- **Causal + anatomical structure** → realistic, faithful radiographs.
- Synthetic augmentation **improves segmentation**, especially for rare classes.
- **Next:** counterfactual & disease-progression simulation.